

Tiling-count sequence of the $3 \times n$ rectangle with two opposite corner cells removed: a new recurrence, its interleave identity and parity laws (Lean 4 machine-checked claim of record)

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Abstract

Let $a(n)$ be the number of matchings (monomer–dimer coverings with 1×2 dominoes, the empty one included) of the $3 \times n$ rectangle with the top-right and the bottom-left cell removed: 1, 5, 33, 204, 1266, 7873, 48882, $\dots$. We state and machine-check, in Lean 4 (v4.34.0, mathlib v4.34.0), four theorems about the integer sequences defined by the initial values and the recurrences fitted to this count: the main order-7 recurrence and the interleaving of the two same-coefficient order-7 subfamily recurrences characterise each other, and the parities follow explicit patterns (period 12 modulo 2 for the main sequence, period 6 for the two subsequences). Every proof compiles with no `sorry` and `#print axioms` reports only {`propext`, `Quot.sound`, `Classical.choice`}. Novelty adjudication: no occupying record found (three consistent rounds), value tier *new sequence* / *new recurrence*, not new mathematics; the perfect-matching sub-sequence at even indices is already registered (OEIS A061278) and is not claimed. The combinatorial identification is not kernel-proved and is stated as a conjecture. This note is a priority claim of record, not a full paper.

1 Introduction

Four theorems about three integer sequences defined by initial values and linear recurrences are reported here; the sequences come from counting match-

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ings of the $3 \times n$ rectangle with the top-right and bottom-left cells removed, and the recurrences were obtained by exact rational fitting of enumerated data. The proofs were produced by the AI4Math pipeline, in which LLM workers generate statements and proof scripts and the Lean 4 kernel is the sole adjudicator at every gate: statements are frozen by hash before proving, and a separate audit re-computes the symmetric difference, scans the whole file for `sorry/admit` including comments, and re-prints the axiom dependencies. Verification strength is stated below verbatim. This note is a priority claim of record deposited so that a dated, citable public record exists ahead of any narrative paper; the fuller paper for the same four theorems is prepared separately and will reference this deposit. No literature review is attempted here.

2 Statement

Throughout, indexing is 0-based in the Lean layer and 1-based in the counting layer: $\text{adiag } n = a(n + 1)$, $\text{onotch } k = a(2k + 1)$ (odd positions) and $\text{enotch } k = a(2k + 2)$ (even positions). The three sequences are *defined* by the displayed initial values and recurrences; the statement layer contains no graphs, no matchings and no bijections (the type is $\mathbb{N} \rightarrow \mathbb{Z}$ throughout).

Listing 1: Sequence `adiag` (main sequence), verbatim from the frozen statement file `formalized/01-comb07-diagnotch-statements.lean` (lines 39–53, sha256 16744fd3).

```

1  /-- **序列 adiag (主列, 七阶递推) **.
2  初值  $a(0..6) = 1, 5, 33, 204, 1266, 7873, 48882$  (0-based; 对应 1-
   based  $a(1..7)$ ) ;
3  递推  $a(n+7) = 4 \cdot a(n+6) + 15 \cdot a(n+5) - 5 \cdot a(n+4) - 19 \cdot a(n+3)$ 
   +  $9 \cdot a(n+2) + 2 \cdot a(n+1) - a(n)$  ( $n \geq 0$ ) 。 -/
4
5  def adiag :  $\mathbb{N} \rightarrow \mathbb{Z}$ 
6  | 0 => 1
7  | 1 => 5
8  | 2 => 33
9  | 3 => 204
10 | 4 => 1266
11 | 5 => 7873
12 | 6 => 48882
13 | n + 7 =>
14   4 * adiag (n + 6) + 15 * adiag (n + 5) - 5 * adiag (n + 4)
15   - 19 * adiag (n + 3) + 9 * adiag (n + 2) + 2 * adiag (n +
   1) - adiag n

```

Listing 2: Sequences `onotch` and `enotch` (the two subsequences, same coefficients), verbatim from the frozen statement file (lines 55–87, sha256 16744fd3).

```

1  /-- **序列 onotch (1-based 奇位子列, 与 enotch 同系数的七阶递推) **
2  。
3  初值  $o(0..6) = 1, 33, 1266, 48882, 1886600, 72804880, 2809532937$ 
4  (= adiag 在下标 0,2,4,6,8,10,12 处之值);
5  递推  $o(k+7) = 46 \cdot o(k+6) - 303 \cdot o(k+5) + 671 \cdot o(k+4) - 519 \cdot o(k+3)$ 
6       $+ 167 \cdot o(k+2) - 22 \cdot o(k+1) + o(k) \ (k \geq 0)$  。 -/
7  def onotch :  $\mathbb{N} \rightarrow \mathbb{Z}$ 
8  | 0 => 1
9  | 1 => 33
10 | 2 => 1266
11 | 3 => 48882
12 | 4 => 1886600
13 | 5 => 72804880
14 | 6 => 2809532937
15 | k + 7 =>
16     46 * onotch (k + 6) - 303 * onotch (k + 5) + 671 * onotch (k
17     + 4)
18     - 519 * onotch (k + 3) + 167 * onotch (k + 2) - 22 *
19     onotch (k + 1) + onotch k
20
21 /-- **序列 enotch (1-based 偶位子列, 与 onotch 同系数的七阶递推) **
22 。
23 初值  $e(0..6) = 5, 204, 7873, 303723, 11720017, 452270456,$ 
24      $17453021829$ 
25 (= adiag 在下标 1,3,5,7,9,11,13 处之值);
26 递推  $e(k+7) = 46 \cdot e(k+6) - 303 \cdot e(k+5) + 671 \cdot e(k+4) - 519 \cdot e(k+3)$ 
27      $+ 167 \cdot e(k+2) - 22 \cdot e(k+1) + e(k) \ (k \geq 0)$  。 -/
28 def enotch :  $\mathbb{N} \rightarrow \mathbb{Z}$ 
29 | 0 => 5
30 | 1 => 204
31 | 2 => 7873
32 | 3 => 303723
33 | 4 => 11720017
34 | 5 => 452270456
35 | 6 => 17453021829
36 | k + 7 =>
37     46 * enotch (k + 6) - 303 * enotch (k + 5) + 671 * enotch (k
38     + 4)
39     - 519 * enotch (k + 3) + 167 * enotch (k + 2) - 22 *
40     enotch (k + 1) + enotch k

```

Theorem 1 (interleaving identity). *Let adiag , onotch , enotch be defined by the initial values and the recurrences of Listings 1 and 2. Then for every $k \in \mathbb{N}$,*

$$\mathit{adiag}(2k) = \mathit{onotch} k \quad \text{and} \quad \mathit{adiag}(2k+1) = \mathit{enotch} k.$$

Equivalently: the full order-7 recurrence and the interleaving of the two same-coefficient order-7 subfamily recurrences characterise each other.

Listing 3: Theorem 1, verbatim from the frozen statement file (lines 89–94, sha256 16744fd3); the listing stops at the theorem’s `:= by` line.

```
1 /-- **T1 (主攻.交织恒等式) **: 对一切  $k$ ,  $\mathit{adiag}(2k) = \mathit{onotch} k$  且
2  $\mathit{adiag}(2k+1) = \mathit{enotch} k$ 。即: 主列的完整七阶递推 = 两条同系数
3 ( $46, \dots, 303, 671, 519, 167, 22, 1$ ) 七阶子列递推 ( $\mathit{onotch}/\mathit{enotch}$ ) 的交
   织,
4 两者互相刻画 ( $0$ -based; 对应  $1$ -based 的  $a(2k+1)=o(k)$ 、 $a(2k+2)=e(k)$ 
   )。 -/
5 theorem  $\mathit{adiag\_interleave}$  ( $k : \mathbb{N}$ ) :
6    $\mathit{adiag}(2 * k) = \mathit{onotch} k \wedge \mathit{adiag}(2 * k + 1) = \mathit{enotch} k := \mathbf{by}$ 
```

Theorem 2 (main sequence, period 12 modulo 2). *For all $n, r \in \mathbb{N}$ with $n \bmod 12 = r$,*

$$\mathit{adiag} n \bmod 2 = \mathit{pat2} r,$$

where $\mathit{pat2}$ is the twelve-entry pattern $(1, 1, 1, 0, 0, 1, 0, 1, 0, 1, 0, 0)$ of Listing 4. Equivalently, $\mathit{adiag} n$ is odd if and only if $n \equiv 0, 1, 2, 5, 7, 9 \pmod{12}$.

Listing 4: Pattern table $\mathit{pat2}$ and Theorem 2, verbatim from the frozen statement file (lines 97–116, sha256 16744fd3); the listing stops at the theorem’s `:= by` line.

```
1 /-- **余数样式  $\mathit{pat2}$ ** ( $12$  留数类  $\mapsto \bmod 2$  余数) :
2  $0 \mapsto 1, 1 \mapsto 1, 2 \mapsto 1, 3 \mapsto 0, 4 \mapsto 0, 5 \mapsto 1, 6 \mapsto 0, 7 \mapsto 1, 8 \mapsto 0, 9 \mapsto 1, 10 \mapsto 0,$ 
    $11 \mapsto 0$ 。 -/
3 def  $\mathit{pat2} : \mathbb{N} \rightarrow \mathbb{Z}$ 
4   | 0 => 1
5   | 1 => 1
6   | 2 => 1
7   | 3 => 0
8   | 4 => 0
9   | 5 => 1
10  | 6 => 0
```

```

11 | 7 => 1
12 | 8 => 0
13 | 9 => 1
14 | 10 => 0
15 | _ => 0
16
17 /-- **T2 (伴随主列 mod 2 周期 12 显式样式) **: adia $n$  的 mod 2 余
    数仅依赖
18  $n \bmod 12$ , 样式为  $pat2$  (即  $adia $n$  为奇  $\iff n \equiv 0, 1, 2, 5, 7, 9 \pmod{12}$ )$ ). -/
19 theorem adia $_{mod2\_period12}$  (n r :  $\mathbb{N}$ ) (hr : n % 12 = r) :
20   adia $n$  % 2 = pat2 r := by

```

Theorem 3 (odd-position subsequence, period 6 modulo 2). *For all $k, r \in \mathbb{N}$ with $k \bmod 6 = r$,*

$$\text{onotch } k \bmod 2 = \text{pat6o } r,$$

where pat6o is the six-entry pattern $(1, 1, 0, 0, 0, 0)$ of Listing 5. Equivalently, $\text{onotch } k$ is odd if and only if $k \equiv 0$ or $1 \pmod{6}$.

Theorem 4 (even-position subsequence, period 6 modulo 2). *For all $k, r \in \mathbb{N}$ with $k \bmod 6 = r$,*

$$\text{enotch } k \bmod 2 = \text{pat6e } r,$$

where pat6e maps 1 and 5 to 0 and every other residue to 1 (Listing 5). Equivalently, $\text{enotch } k$ is even if and only if $k \equiv 1$ or $5 \pmod{6}$.

Listing 5: Pattern tables pat6o and pat6e , verbatim from the frozen statement file (lines 119–131, sha256 16744fd3).

```

1 /-- **余数样式 pat6o** (onotch 侧 6 留数类  $\mapsto$  mod 2 余数) :
2    $0 \mapsto 1, 1 \mapsto 1$ , 其余  $\mapsto 0$  (即  $\text{onotch } k$  为奇  $\iff k \equiv 0$  或  $1 \pmod{6}$ ) .
3   -/
4   def pat6o :  $\mathbb{N} \rightarrow \mathbb{Z}$ 
5     | 0 => 1
6     | 1 => 1
7     | _ => 0
8
9 /-- **余数样式 pat6e** (enotch 侧 6 留数类  $\mapsto$  mod 2 余数) :
10   $1 \mapsto 0, 5 \mapsto 0$ , 其余  $\mapsto 1$  (即  $\text{enotch } k$  为偶  $\iff k \equiv 1$  或  $5 \pmod{6}$ ) .
11  -/
12  def pat6e :  $\mathbb{N} \rightarrow \mathbb{Z}$ 
13    | 1 => 0

```

```

12 | 5 => 0
13 | _ => 1

```

Listing 6: Theorem 3, verbatim from the frozen statement file (lines 133–136, sha256 16744fd3); the listing stops at the theorem’s `:= by` line.

```

1 /-- **T3 之 onotch (伴随·奇位子列 mod 2 周期 6 样式) **: onotch k 的
   mod 2 余数
2 仅依赖 k mod 6, 样式为 pat6o (即 onotch k 为奇  $\iff k \equiv 0$  或  $1 \pmod{6}$ ) )。 -/
3 theorem onotch_mod2_period6 (k r : ℕ) (hr : k % 6 = r) :
4   onotch k % 2 = pat6o r := by

```

Listing 7: Theorem 4, verbatim from the frozen statement file (lines 139–142, sha256 16744fd3); the listing stops at the theorem’s `:= by` line.

```

1 /-- **T3 之 enotch (伴随·偶位子列 mod 2 周期 6 样式) **: enotch k 的
   mod 2 余数
2 仅依赖 k mod 6, 样式为 pat6e (即 enotch k 为偶  $\iff k \equiv 1$  或  $5 \pmod{6}$ ) )。 -/
3 theorem enotch_mod2_period6 (k r : ℕ) (hr : k % 6 = r) :
4   enotch k % 2 = pat6e r := by

```

Conjecture 1 (combinatorial bridge, not proved here). For every $n \geq 1$, the matching count $a(n)$ of the $3 \times n$ rectangle with the top-right and bottom-left cells removed equals $\text{adiag}(n-1)$; equivalently, $\text{onotch } k = a(2k+1)$ and $\text{enotch } k = a(2k+2)$ for every k . This identification is not kernel-proved, is not used by any theorem of this note, and rests on exact enumeration by independently written programs with different coverage.

3 Proof

All four proofs are strong inductions on the index, with the seed window discharged by computation (`decide`) and the step closed by congruence algebra. Theorem 1 first proves an auxiliary 14-step identity (`lifted`) by strong induction: for $n < 7$ both sides are explicit integers; for $n = m + 7$ the definitional recurrence is unfolded at the eight indices $m+7, \dots, m+21$, the seven induction hypotheses are rewritten into the same 14-step form, and a single `linear_combination` of these fifteen equations closes the step by cancellation. The main statement is then a strong induction on k : for $k < 7$ the fourteen equalities are concrete integer identities; for $k = j + 7$

the even half applies `lifted` at $2j$, normalises the indices by `ring`, replaces the seven predecessors by the induction hypotheses, and closes by the definitional unfolding of `onotch` (the odd half is identical, ending in `enotch`). Theorems 2–4 share one skeleton: strong induction, one definitional unfolding whose coefficients collapse modulo 2 to 0, 1, 1, 1, 1, 0, 1, and a finite check per residue class (`mod_cases` with `omega/decide`).

The complete proof file has 433 lines and therefore exceeds one page; the four contiguous verbatim excerpts below carry the distinctive argument shapes, and the complete file ships in this deposit under `proofs/`.

Listing 8: Inductive step of the auxiliary lemma `lifted`: the last unfolding `p14`, the seven induction hypotheses `h0–h6` and the closing `linear_combination`, verbatim excerpt from `final/01-comb07-diagnotch-proved.lean` (lines 147–172, sha256 51721e9e).

```

1      have p14 : adiag (m + 21) = 4 * adiag (m + 20) + 15 *
      adiag (m + 19)
2      - 5 * adiag (m + 18) - 19 * adiag (m + 17) + 9 * adiag
      (m + 16)
3      + 2 * adiag (m + 15) - adiag (m + 14) := by simp only
      [adiag]
4      have h0 : adiag (m + 14) = 46 * adiag (m + 12) - 303 *
      adiag (m + 10)
5      + 671 * adiag (m + 8) - 519 * adiag (m + 6) + 167 *
      adiag (m + 4)
6      - 22 * adiag (m + 2) + adiag m := ihn m (by omega)
7      have h1 : adiag (m + 15) = 46 * adiag (m + 13) - 303 *
      adiag (m + 11)
8      + 671 * adiag (m + 9) - 519 * adiag (m + 7) + 167 *
      adiag (m + 5)
9      - 22 * adiag (m + 3) + adiag (m + 1) := ihn (m + 1) (
      by omega)
10     have h2 : adiag (m + 16) = 46 * adiag (m + 14) - 303 *
      adiag (m + 12)
11     + 671 * adiag (m + 10) - 519 * adiag (m + 8) + 167 *
      adiag (m + 6)
12     - 22 * adiag (m + 4) + adiag (m + 2) := ihn (m + 2) (
      by omega)
13     have h3 : adiag (m + 17) = 46 * adiag (m + 15) - 303 *
      adiag (m + 13)
14     + 671 * adiag (m + 11) - 519 * adiag (m + 9) + 167 *
      adiag (m + 7)
15     - 22 * adiag (m + 5) + adiag (m + 3) := ihn (m + 3) (

```

```

16   by omega)
17   have h4 : adiaq (m + 18) = 46 * adiaq (m + 16) - 303 *
    adiaq (m + 14)
18     + 671 * adiaq (m + 12) - 519 * adiaq (m + 10) + 167 *
    adiaq (m + 8)
19     - 22 * adiaq (m + 6) + adiaq (m + 4) := ihn (m + 4) (
    by omega)
20   have h5 : adiaq (m + 19) = 46 * adiaq (m + 17) - 303 *
    adiaq (m + 15)
21     + 671 * adiaq (m + 13) - 519 * adiaq (m + 11) + 167 *
    adiaq (m + 9)
22     - 22 * adiaq (m + 7) + adiaq (m + 5) := ihn (m + 5) (
    by omega)
23   have h6 : adiaq (m + 20) = 46 * adiaq (m + 18) - 303 *
    adiaq (m + 16)
24     + 671 * adiaq (m + 14) - 519 * adiaq (m + 12) + 167 *
    adiaq (m + 10)
25     - 22 * adiaq (m + 8) + adiaq (m + 6) := ihn (m + 6) (
    by omega)
26   linear_combination 4 * h6 + 15 * h5 - 5 * h4 - 19 * h3 + 9
    * h2 + 2 * h1 - h0
    + p14 - 46 * p12 + 303 * p10 - 671 * p8 + 519 * p6 - 167
    * p4 + 22 * p2 - p0

```

Listing 9: Even half of the inductive step of Theorem 1, verbatim excerpt from `final/01-comb07-diagnotch-proved.lean` (lines 179–190, sha256 51721e9e).

```

1   · rw [show 2 * (j + 7) = 2 * j + 14 by ring, lifted (2 * j)]
2   rw [show 2 * j + 12 = 2 * (j + 6) by ring,
3     show 2 * j + 10 = 2 * (j + 5) by ring,
4     show 2 * j + 8 = 2 * (j + 4) by ring,
5     show 2 * j + 6 = 2 * (j + 3) by ring,
6     show 2 * j + 4 = 2 * (j + 2) by ring,
7     show 2 * j + 2 = 2 * (j + 1) by ring]
8   rw [(ih (j + 6) (by omega)).1, (ih (j + 5) (by omega)).1,
9     (ih (j + 4) (by omega)).1, (ih (j + 3) (by omega)).1,
10    (ih (j + 2) (by omega)).1, (ih (j + 1) (by omega)).1,
11    (ih j (by omega)).1]
12   simp only [onotch]

```

Listing 10: Core of Theorem 2: the modulo-2 collapse `h1` and the first residue branch, verbatim excerpt from `final/01-comb07-diagnotch-proved.lean`

(lines 215–229, sha256 51721e9e).

```

1      have h1 : adiag (j + 7) % 2
2          = (adiag (j + 5) % 2 + adiag (j + 4) % 2 + adiag (j +
3              3) % 2
4              + adiag (j + 2) % 2 + adiag j % 2) % 2 := by
5              simp only [adiag]
6              omega
7      rw [h1, ih (j + 5) (by omega), ih (j + 4) (by omega),
8          ih (j + 3) (by omega), ih (j + 2) (by omega), ih j (by
9              omega)]
10     mod_cases hj : j % 12
11     · have hj0 : j % 12 = 0 := by simp [Nat.ModEq] using hj
12       have h5 : (j + 5) % 12 = 5 := by omega
13       have h4 : (j + 4) % 12 = 4 := by omega
14       have h3 : (j + 3) % 12 = 3 := by omega
15       have h2 : (j + 2) % 12 = 2 := by omega
16       have h7 : (j + 7) % 12 = 7 := by omega
17       rw [hj0, h5, h4, h3, h2, h7]; decide

```

Listing 11: Core of Theorem 3, verbatim excerpt from final/01-comb07-diagnotch-proved.lean (lines 320–334, sha256 51721e9e); Theorem 4 has the same shape at lines 383–397.

```

1      have h1 : onotch (j + 7) % 2
2          = (onotch (j + 5) % 2 + onotch (j + 4) % 2 + onotch (j
3              + 3) % 2
4              + onotch (j + 2) % 2 + onotch j % 2) % 2 := by
5              simp only [onotch]
6              omega
7      rw [h1, ih (j + 5) (by omega), ih (j + 4) (by omega),
8          ih (j + 3) (by omega), ih (j + 2) (by omega), ih j (by
9              omega)]
10     mod_cases hj : j % 6
11     · have hj0 : j % 6 = 0 := by simp [Nat.ModEq] using hj
12       have h5 : (j + 5) % 6 = 5 := by omega
13       have h4 : (j + 4) % 6 = 4 := by omega
14       have h3 : (j + 3) % 6 = 3 := by omega
15       have h2 : (j + 2) % 6 = 2 := by omega
16       have h7 : (j + 7) % 6 = 1 := by omega
17       rw [hj0, h5, h4, h3, h2, h7]; decide

```

4 Verification evidence

- Toolchain: Lean v4.34.0 [1] (leanprover/lean4:v4.34.0) and mathlib4 v4.34.0, revision 5ed29652 [2], pinned. Reproduction: with that pin, `lake env lean on proofs/01-comb07-diagnotch-proved.lean` compiles the file with exit code 0 (measured 37.0s for the assembled final file in the strict pass).
- Statement integrity: the frozen statement file `formalized/01-comb07-diagnotch-statements.lean` (143 lines) has sha256 16744fd37b7d8eb67965641f4e2b6dd87e3c2a994ea484766b6e3afb5861aca; the final file `final/01-comb07-diagnotch-proved.lean` (433 lines) has sha256 51721e9e315db12a35ca62218d8d458f32443e97f666e4a2ab452d0056f7cb43. The final file was compared with the frozen file by symmetric difference over all ten declarations (six `def` and four `theorem`): zero content difference, only a reordering (the six definitions were moved in front of the theorems); each theorem header matches byte-for-byte up to the `:=` by line.
- Kernel check: compiles with 0 `sorry` (whole-file scan including comments, case-insensitive, also for `admit/sorryAx`); `#print axioms` output, verbatim (kept literal; the smaller monospace width is typesetting only):

```
1 'adiag_interleave' depends on axioms: [propeXt, Classical.choice,
    Quot.sound]
2 'adiag_mod2_period12' depends on axioms: [propeXt, Quot.sound]
3 'onotch_mod2_period6' depends on axioms: [propeXt, Quot.sound]
4 'enotch_mod2_period6' depends on axioms: [propeXt, Quot.sound]
```

The strict re-compile emits no error and no warning. The four axiom lines are the audit department's independent probe output; the gatebatch sidecar files `axioms-adiag_interleave.txt` and its three siblings ship in this deposit under `audit/` with the same four lines each.

- Grading by the audit department: all four theorems *complete proof*; lowest grade reached for the case = *complete proof*; no scaffolding route was used.
- Audit chain (originals in this deposit under `audit/`): final adjudication `final-comb07-diagnotch.txt` (four independent checks, including the symmetric difference and the independent axiom probe),

the four axiom printouts `axioms-adiag_interleave.txt`, `axioms-adiag_mod2_period12.txt`, `axioms-onotch_mod2_period6.txt`, `axioms-enotch_mod2_period6.txt`, and the full C9 query log `c9-record.md`. The semantic-fidelity check (roundtrip re-translation) of the underlying task ran on the same model node as its formalisation; that independence weakness is mitigated, not removed, by kernel adjudication and manual review.

5 Novelty evidence

Adjudication copied from the pipeline’s exclusivity review (C9 gate), which ran three rounds before the problem was accepted, by workers who re-derived the data independently rather than trusting any external report: verdict **no occupying record found** (three consistent rounds), value tier **new sequence / new recurrence**. The tier is not a claim of new mathematics and not a first-discovery claim: the order-7 recurrence is machine-verified to be of a different spectrum than the order-6 recurrence of the undefected mother family A033506 (neither satisfies the other’s recurrence; the two characteristic polynomials share one quintic factor), and the counting method itself (state-matrix recursion) is documented prior art that would produce this sequence as a special case. Summary by channel; the full query log, including every zero-hit query, is in this deposit at `audit/c9-record.md`.

channel	queries	result
OEIS sequence search [6]	OEIS independent strings (rounds 1-3)	zero hit; full string, problem, table, >2 characteristic polynomials and enumerative base strings, word-3 subsequences; anchors self-verified
OEIS pre-family records	2	full-column reads of A033506/A033507 [4, 5]; references and links cover complete grids only, no defect variant
OpenMath	7	zero same-form (top hits: dinner physics / knowledge graphs / materials science)
AMATE	5	zero same-form
arXiv	5	zero same-form (total=0 on two alias groups)
Crossref	3	zero same-form
Os 2019, nearest neighbour	1	[5]: Theorem 4 (fixed measure set) is a method with a different statistic; no instance of this sequence; reference chain (Kong 2006, Teng, Wu 2003, Wu 2006) and 12 citing works read, none enumerates defective-grid all-matchings
Lean enumerators	2 + tree walk	zero hit; match has no matching-count theorems; compile zero sequence with structurally absent public Lean code search on site large terms: total=0

Two boundaries are explicit. The perfect-matching sub-sequence at even indices (1, 5, 20, 76, 285, 1065, . . .) is already registered as OEIS A061278 [7] and is not claimed as new; the enumeration anchors A033507 and A030186 [5, 6] are likewise registry entries, not claims. And the residual gaps, kept from the audit trail: the general web search layer was unavailable in the review environment, so the literature evidence rests on the four structured channels above; Lundow 1996/1998 and Read 1982 (the mother-family founding literature) were not read in full text; Guichard 2008 is behind a paywall; and the Math Stack Exchange API was rate-limited. If any of these channels later returns a hit on the defective-grid matching counts, the tier above weakens accordingly, and this deposit will be amended by a later version rather than by editing this record.

6 Scope and limitations

Grade and boundary, stated as the audit states them. (a) The theorems cover the recurrence-defined sequences only; their equality with the matching counts of the notched rectangle is not kernel-proved and is Conjecture 1. Its empirical support is enumeration by exact programs: the main counting program covers $1 \leq n \leq 50$, cross-checked against a delete-and-branch enumerator for $1 \leq n \leq 14$ and against a brute-force backtracker for $1 \leq n \leq 9$, with the undefected $3 \times n$, $4 \times n$ and $2 \times n$ families reproduced as external anchors; the enumerators themselves are not verified, and the scripts and their outputs stay in the working repository under `tasks/20260928-comb07-diagnotch/`. (b) The recurrences are computational findings: Berlekamp–Massey fits over $\mathbb{Q}$ with minimal order exactly 7, verified digit-for-digit against enumeration, not derived from a transfer-matrix argument. (c) The theorems are about parity only; the mod-4 pattern (period 12), the mod-3 patterns (periods 968 and 484) and a general periodicity statement were examined during claim selection and deliberately left unproved in this batch, and appear in no theorem here. (d) The perfect-matching sub-sequence at even indices is already registered as OEIS A061278 and is excluded from the novelty claim; the value tier “new sequence / new recurrence” carries only the spectral-difference meaning recorded above. (e) The gate-four sign-off of the underlying task was exercised by the author’s instruction to run this claim lane on 2026-09-28, while the task report file still carries its draft header (pending manual sign-off at gate four); both facts are recorded here as they stand, and the task products themselves were not edited. (f) This note carries no literature review and is a deposit record, not a survey; the narrative paper for the same result is separate.

Data availability

The deposit package for this claim contains the claim note (LaTeX source and PDF), the Lean sources (`proofs/`: frozen statement file, final proof file, `lean-toolchain`), and the verification and novelty-search records (`audit/`). Zenodo: the DOI is reserved at upload by the author and is recorded in `metadata/README.md` of this package once created; at the time of writing it is an unfilled field, `RESERVED-DOI`, not a value (see the upload steps in `UPLOAD.md` in the source repository, which is not shipped with this deposit). Public mirror of this deposit: the claim lane’s repository <https://github.com/Aurora0134/ai4math-claims>,

directory `grid3n-diag-2notch/` (pushed under the author’s instruction of 2026-09-28; the commit and timestamp are recorded in the source repository’s `claims/grid3n-diag-2notch/card.md`). Claim note under CC BY 4.0, code under MIT.

Use of generative AI

(a) AI performed: candidate-problem screening and the three-round exclusivity review, the numerical recomputation and claim selection of phase 0, autoformalisation of the four statements and the roundtrip semantic check, proof search and proof-script writing, the prose translation of the proofs, and the assembly of this note including its verbatim listings. (b) Model, node and budget: all task-stage sampling ran on a single node, wire-id `anthropic/a6api-main/kimi-k3`, with no cross-node switching. One drift is disclosed: one audit dispatch on 2026-09-28 (the F1 closure re-review) carried a model-detection reading of `anthropic/stepfun/step-5-preview`; the ledger’s accounting node stayed `anthropic/a6api-main/kimi-k3` and the drift is disclosed rather than reconciled by substitution. The task consumed 11 agent-run dispatches of a permitted 20 and 1 `llm-call` of a permitted 6 (the roundtrip semantic judge; two earlier network failures on the same node were retried and not counted), numbers copied verbatim from the task ledger `tasks/20260928-comb07-diagnotch/budget.log` (a source-repository file, not shipped with this package). This note consumed 0 pipeline `llm-call` invocations and 1 agent-run dispatch (this assembly) against caps of 4 and 4 respectively, and its LaTeX compilation used the rounds recorded in `audit/compile.txt`; the assembly session’s endpoint was `zcode`, whose model-detection reading of 2026-09-28 resolved to `stepfun/step-5-preview`, which is disclosed rather than replaced by the task node’s name. (c) Final adjudication: all statements and proofs reported here were checked by the Lean 4 kernel: every proof compiles with no `sorry`, and `#print axioms` reports only `{propext, Quot.sound, Classical.choice}` — exactly this set for `adiag_interleave`, and `{propext, Quot.sound}` for `adiag_mod2_period12`, `onotch_mod2_period6` and `enotch_mod2_period6`. (d) The AI system is not an author; the human author reviewed the evidence and is responsible for all content.

Author contributions

CRediT-style: the human author adjudicated topic selection, statement semantics, the sign-off and the deposit decision; the AI4Math pipeline (an LLM) generated the candidate propositions, the proof searches and the text drafts. The AI system is not listed as an author.

References

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- [6] The OEIS Foundation. Entry A030186: number of matchings in graph $P_2 \times P_n$ ($a(n) = 3a(n-1) + a(n-2) - a(n-3)$). *The On-Line Encyclopedia of Integer Sequences*. <https://oeis.org/A030186>
- [7] The OEIS Foundation. Entry A061278: $a(n) = 5a(n-1) - 5a(n-2) + a(n-3)$ — the perfect-matching sub-sequence at even n of this note’s graph family, already registered and not claimed as new. *The On-Line Encyclopedia of Integer Sequences*. <https://oeis.org/A061278>
- [8] The OEIS Foundation. *The On-Line Encyclopedia of Integer Sequences*. <https://oeis.org/> — searched for the sequences, their residue-class subsequences, their characteristic polynomials and their generating-function numerators, with no entry found (see Section 5).